

Curriculum Vitae: updated September 13, 2026

I. IDENTIFYING DATA

Name M. Brandon Westover, MD, PhD
Current Position Emily Fisher Landau Professor of Neurology, Harvard Medical School, Beth Israel Deaconess Medical Center (BIDMC)
Current Affiliation Beth Israel Deaconess Medical Center (BIDMC)

II. EDUCATION HISTORY

Colleges and Universities Attended

7/1999 BS in Physics, Brigham Young University
7/2006 PhD, Physics, Washington University in St. Louis
7/2006 MD, Washington University School of Medicine

Residency and Fellowship Training

2006 – 2007 Internship, Barnes Jewish Hospital, Washington University School of Medicine, St. Louis, MO
2007 – 2010 Residency, Partners Neurology Program (Massachusetts General Hospital and Brigham and Women's Hospital), Boston, MA
2010 – 2012 Fellowship, Epilepsy and Clinical Neurophysiology, Massachusetts General Hospital, Boston, MA

Board Certification

2011 American Board of Psychiatry and Neurology (No. 56586)

III. EMPLOYMENT

Academic Appointments:

07/12-05/14 Instructor in Neurology, Harvard Medical School, Boston, MA
05/14-05/18 Assistant Professor in Neurology, Harvard Medical School, Boston, MA
05/18-10/22 Associate Professor
10/22 – present Professor of Neurology, Harvard Medical School, Boston, MA
11/22 – present Emily Fisher Landau Professor of Neurology, Harvard Medical School, Boston, MA

Other Appointments:

2022 to present Director, Critical Care EEG Monitoring Service, BIDMC

IV. HONORS AND AWARDS

2011 Best Paper Award Statistical Literacy (www.statlit.org) Statistical Education
2012 Best Paper Award Association for Computing Machinery: Special Interest Group on Knowledge Discovery and Data Mining (SIGKDD) Basic Research
2014 Rappaport Fellowship Massachusetts General Hospital Nominated award
2018 #1 Most Disruptive AI Innovator Partners Healthcare Partners World Innovation Forum
2018 Innovation Discovery Grant Partners Innovation Office Artificial Intelligence Research / Development
2018 MGH-MIT Grand Challenge Winner ECOR Artificial Intelligence Research
2018 Best Neurology Clinical Research Abstract MGH Neurology Department Clinical Research
2018 Best Paper Award CyberWorlds 2018, Singapore Computer Science

2019	Disruptive Dozen AI Innovator	Massachusetts General Hospital Partners HealthcareWorld Innovation Forum
2019	Innovation Discovery Award (IDG)	Partners Healthcare World Innovation Forum
2019	Innovative Applications in Analytics Award (IAAA)	INFORMS Conference on Business Analytics & Operations ResearchFor achievements featuring novel combinations of descriptive, predictive and prescriptive analytics.
2019	Best Oral Abstract: Translational American Delirium Society (ADS)	ADS annual meeting
2019	Derek Denny-Brown Young Neurological Scholar Award	American Neurological Association
	ANA's Highest award. For outstanding scientific advances toward the prevention, diagnosis, treatment, and cure of neurological diseases	

V. BIBLIOGRAPHY

Peer-reviewed original research (370 total)

1. Eliasmith C, **Westover MB**, Anderson CH. "A general framework for neurobiological modeling: an application to the vestibular system." *Neurocomputing* (Amst). 2002 Jun;44-46:1071-6. doi: 10.1016/s0925-2312(02)00418-6. PMID: PMC6788744.
2. **Westover MB**, Eliasmith C, Anderson CH. "Linearly decodable functions from neural population codes." *Neurocomputing* (Amst). 2002 Jun;44-46:691-696. doi: 10.1016/s0925-2312(02)00459-9. Epub 2002 Mar 26. PMID: PMC7062372.
3. Fischer BJ, **Westover MB**. "The neural multiple access channel." *Neurocomputing* (Amst). 2003 Jun;52-54:511-518. doi: 10.1016/s0925-2312(02)00762-2. Epub 2003 Jan 29. PMID: PMC7062373.
4. **Westover MB**, Anderson CH. "Layer 4C in monkey V1 may linearize the output of the LGN." *Neurocomputing* (Amst). 2003 Jun;52-54:671-676. doi: 10.1016/s0925-2312(02)00863-9. Epub 2003 Jan 28. PMID: PMC7062374.
5. **Westover MB**, O'Sullivan JA. "Achievable Rates for Pattern Recognition." *IEEE Trans Inf Theory*. 2008 Jan;54(1):299-320. doi: 10.1109/tit.2007.911296. Epub 2008 Jan 04. PMID: PMC7062371.
6. **Westover MB**. "Asymptotic Geometry of Multiple Hypothesis Testing." *IEEE Trans Inf Theory*. 2008 Jul;54(7):3327-3329. doi: 10.1109/TIT.2008.924656. Epub 2008 Jun 17. PMID: PMC6788803.
7. Woeltje KF, Butler AM, Goris AJ, Tutlam NT, Doherty JA, **Westover MB**, Ferris V, Bailey TC. "Automated surveillance for central line-associated bloodstream infection in intensive care units." *Infect Control Hosp Epidemiol*. 2008 Sep;29(9):842-6. doi: 10.1086/590261. PMID: PMC6788749.
8. Mueen A, Keogh E, Zhu Q, Cash S, **Westover B**. "Exact Discovery of Time Series Motifs." *Proc SIAM Int Conf Data Min*. 2009;2009:473-484. doi: 10.1137/1.9781611972795.41. PMID: PMC6814436.
9. Chu-Shore J, **Westover MB**, Bianchi MT. "Power law versus exponential state transition dynamics: application to sleep-wake architecture." *PLoS One*. 2010 Dec 02;5(12):e14204. doi: 10.1371/journal.pone.0014204. PMID: PMC2996311.
10. Mueen A, Keogh E, Zhu Q, Cash SS, **Westover MB**, Bigdely-Shamlo N. "A disk-aware algorithm for time series motif discovery." *Data Min Knowl Discov*. 2011 Jan;22(1-2):73-105. doi: 10.1007/s10618-010-0176-8. Epub 2010 Apr 18. PMID: PMC7062370.
11. Zhang WJ, **Westover MB**, Keary CJ. "Premortem diagnosis of sporadic Creutzfeldt-Jakob disease aided by positron-emission tomography imaging." *AJNR Am J Neuroradiol*. 2011 Jan;32(1):E18. doi: 10.3174/ajnr.A2292. Epub 2010 Nov 11. PMID: PMC6788750.
12. **Westover MB**, Bianchi MT, Eckman MH, Greenberg SM. "Statin use following intracerebral hemorrhage: a decision analysis." *Arch Neurol*. 2011 May;68(5):573-9. doi: 10.1001/archneurol.2010.356. Epub 2011 Jan 10. PMID: PMC3158138.
 - Featured article with accompanying editorial: *Arch Neurol*. 2011;68(5):565-566. "Statins After Intracerebral Hemorrhage: To Treat or Not to Treat".
 - Covered in over 150 news feeds, including CBS News, ABC News, US News and World Report, and Medscape Medical News.

13. **Westover MB, Westover KD**, Bianchi MT. "Significance testing as perverse probabilistic reasoning." BMC Med. 2011 Feb 28;9:20. doi: 10.1186/1741-7015-9-20. PMCID: PMC3058025.
14. **Westover KD, Westover MB**, Winer EP, Richardson AL, Iglehart JD, Punglia RS. "Should a sentinel node biopsy be performed in patients with high-risk breast cancer?." Int J Breast Cancer. 2011;2011:973245. doi: 10.4061/2011/973245. Epub 2011 Aug 10. PMCID: PMC3262582.
15. Eiseman NA, **Westover MB**, Mietus JE, Thomas RJ, Bianchi MT. "Classification algorithms for predicting sleepiness and sleep apnea severity." J Sleep Res. 2012 Feb;21(1):101-12. doi: 10.1111/j.1365-2869.2011.00935.x. Epub 2011 Jul 14. PMCID: PMC3698244.
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22. Rakthanmanon T, Campana B, Mueen A, Batista G, **Westover B**, Zhu Q, Zakaria J, Keogh E. "Searching and Mining Trillions of Time Series Subsequences under Dynamic Time Warping." KDD. 2012 Aug;2012:262-270. doi: 10.1145/2339530.2339576. PMCID: PMC6816304.
 • Best Paper Award at Association for Computing Machinery's annual SIGKDD (Special Interest Group on Knowledge Discovery and Data Mining) meeting, 2012.
23. Eiseman NA, **Westover MB**, Ellenbogen JM, Bianchi MT. "The impact of body posture and sleep stages on sleep apnea severity in adults." J Clin Sleep Med. 2012 Dec 15;8(6):655-66A. doi: 10.5664/jcsm.2258. PMCID: PMC3501662.
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Westover MB, Eiseman NA, Cash SS, Bianchi MT. "Information theoretic quantification of diagnostic uncertainty." Open Med Inform J. 2012;6:36-50. doi: 10.2174/1874431101206010036. Epub 2012 Dec 14. PMCID: PMC3537080.
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32. **Westover MB**, Bianchi MT, Yang C, Schneider JA, Greenberg SM. "Estimating cerebral microinfarct burden from autopsy samples." *Neurology*. 2013 Apr 09;80(15):1365-9. doi: 10.1212/WNL.0b013e31828c2f52. Epub 2013 Mar 13. PMID: PMC3662273.
33. Featured article with accompanying editorial: *Neurology*. 2013 Apr 9;80(15):1358-9, "Cerebral Microinfarcts: Enumerating the Innumerable".
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 - Brandon **Westover M**, Shafi MM, Ching S, Chemali JJ, Purdon PL, Cash SS, Brown EN. "Real-time segmentation of burst suppression patterns in critical care EEG monitoring." *J Neurosci Methods*. 2013 Sep 30;219(1):131-41. doi: 10.1016/j.jneumeth.2013.07.003. Epub 2013 Jul 23. PMID: PMC3939433.
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 - Berkowitz AL, **Westover MB**, Bianchi MT, Chou SH. "Aspirin for acute stroke of unknown etiology in

- resource-limited settings: a decision analysis.” *Neurology*. 2014 Aug 26;83(9):787-93. doi: 10.1212/WNL.0000000000000730. Epub 2014 Jul 23. PMID: PMC4155044.
- Gaspard N, Hirsch LJ, LaRoche SM, Hahn CD, **Westover MB**, Critical Care EEG Monitoring Research Consortium. “Interrater agreement for Critical Care EEG Terminology.” *Epilepsia*. 2014 Sep;55(9):1366-73. doi: 10.1111/epi.12653. Epub 2014 Jun 02. PMID: PMC4879939.
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 - Bianchi MT, Lipoma T, Darling C, Alameddine Y, **Westover MB**. “Automated sleep apnea quantification based on respiratory movement.” *Int J Med Sci*. 2014;11(8):796-802. doi: 10.7150/ijms.9303. Epub 2014 May 30. PMID: PMC4057486.
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 46. Akeju O, **Westover MB**, Pavone KJ, Sampson AL, Hartnack KE, Brown EN, Purdon PL. “Effects of sevoflurane and propofol on frontal electroencephalogram power and coherence.” *Anesthesiology*. 2014 Nov;121(5):990-8. doi: 10.1097/ALN.0000000000000436. PMID: PMC4206606.
 47. Akama-Garren EH, Bianchi MT, Leveroni C, Cole AJ, Cash SS, **Westover MB**. “Weighing the value of memory loss in the surgical evaluation of left temporal lobe epilepsy: a decision analysis.” *Epilepsia*. 2014 Nov;55(11):1844-53. doi: 10.1111/epi.12790. Epub 2014 Sep 19. PMID: PMC4877127.
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 50. **Westover MB**, Shafi MM, Bianchi MT, Moura LM, O'Rourke D, Rosenthal ES, Chu CJ, Donovan S, Hoch DB, Kilbride RD, Cole AJ, Cash SS. “The probability of seizures during EEG monitoring in critically ill adults.” *Clin Neurophysiol*. 2015 Mar;126(3):463-71. doi: 10.1016/j.clinph.2014.05.037. Epub 2014 Jul 11. PMID: PMC4289643.
 51. Featured article with accompanying editorial: “Early epileptiform discharges and the yield of prolonged EEG monitoring.” Dangayach N, Claassen J., *Clinical Neurophysiology* (2014), doi: <http://dx.doi.org/10.1016/j.clinph.2014.07.002>.
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 54. Kenny JD, **Westover MB**, Ching S, Brown EN, Solt K. “Propofol and sevoflurane induce distinct burst suppression patterns in rats.” *Front Syst Neurosci*. 2014;8:237. doi: 10.3389/fnsys.2014.00237. Epub 2014 Dec 18. PMID: PMC4270179.
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 56. de Gusmão CM, Berkowitz AL, Hung AY, **Westover MB**. “Cerebrospinal fluid shunt-induced chorea: case report and review of the literature on shunt-related movement disorders.” *Pract Neurol*. 2015

- Feb;15(1):42-4. doi: 10.1136/practneurol-2014-000913. Epub 2014 Jul 04. PMID: PMC4870013.
57. Shafi MM, Vernet M, Klooster D, Chu CJ, Boric K, Barnard ME, Romatoski K, **Westover MB**, Christodoulou JA, Gabrieli JD, Whitfield-Gabrieli S, Pascual-Leone A, Chang BS. "Physiological consequences of abnormal connectivity in a developmental epilepsy." *Ann Neurol*. 2015 Mar;77(3):487-503. doi: 10.1002/ana.24343. Epub 2015 Jan 29. PMID: PMC4394240.
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Peer-reviewed publications (accepted, in press)

1. Nascimento FA, Katyal R, Kass NR, Yuan D, Sirven JI, Westover MB, Beniczky S. "Electroencephalography in clinical practice: results from an AAN Practice Current survey." *Journal of Clinical Neurophysiology*.**
2. Sartipi S, Godfrey DS, Tauţan AM, Fernandes MP, Ghanta M, Gupta A, Nearing B, Kim J, Struck AF, Loddenkemper T, Peters J, Lee JW, Westover MB, Zafar SF. Automated Extraction of Seizures and Ictal-Interictal Continuum Patterns from EEG Reports to Enable Large-Scale Neurophysiology and Neurocritical Care Research. Available at SSRN. doi: 10.2139/ssrn.5489575. SSRN.**

Peer-reviewed publications (other - 8 total)

1. Westover MB. "Review of Peter Dayan & L. F. Abbott's Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems." *Philosophical Psychology*. 2002. 15: 563-567.
2. Bianchi MT and Westover MB. (2009) "Comment on: Analysis of SPARCL trial data for reduction of future stroke risk in older stroke patients with recent stroke or TIA." *Neurology* 73(10):817-818.
3. Westover MB, Gaspard N. "Not a simple plumbing problem: Updating our understanding of delayed cerebral ischemia in aneurysmal subarachnoid hemorrhage." *J Clin Neurophysiol*. 2016 Jun;33(3):171-3. PMCID: PMC4894327. Invited review as Guest Editor for a special issue of JCN on the neurophysiology of subarachnoid hemorrhage.
4. Dreier J, Fabricius M, Cenk A, Sakowitz O, Shuttleworth C, Dohmen C, Graf R, Vajkoczy P, Helbok R, Suzuki M; Schiefecker A.; Major S, Winkler M, Kang EJ, Milakara D, Oliveira-Ferreira A, Reiffurth C, Revankar G, Sugimoto K, Dengler N, Hecht N, Foreman B, Feyen B, Kondziella D, Friberg C, Piilgaard H, Rosenthal ES, Westover MB, Maslarova A, Santos E, Hertle D, Sanchez-Porrás R, Jewell S, Balança B, Platz J, Hinzman J, Luckl J, Schoknecht K, Schöll M, Drenckhahn C, Feuerstein D, Eriksen N, Horst V, Bretz J, Jahnke P, Scheel M, Böhner G, Rostrup E, Pakkenberg B, Heinemann U, Claassen J, Carlson A, Kowoll C, Lublinsky S, Chassidim Y, Shelef I, Friedman A, Brinker G, Reiner M, Kirov S, Andrew R, Farkas E, Guerresir, E, Vatter H, Chung L, Brennan K, Lieutaud T, Marinesco S, Maas A, Sahuquillo J, Dahlem M, Richter F, Herreras O, Boutelle M, Okonkwo D, Bullock R, Witte OW, Martus P, Van den Maagdenberg A, Ferrari M, Dijkhuizen R, Shutter L, Andaluz N, Schulte A, MacVicar B, Watanabe T, Woitzik J, Lauritzen M, Strong A, Hartings J. "Recording, analysis, and interpretation of spreading depolarizations in neurointensive care: review and recommendations of the COSBID research group." *Journal of Cerebral Blood Flow and Metabolism*. 2016 Jun 17. pii: 0271678X16654496. [Epub ahead of print] PMID: 27317657.
5. Westover MB, Halford JJ, Bianchi MT. What it should mean for an algorithm to pass a statistical Turing test for detection of epileptiform discharges. *Clin Neurophysiol*. 2017 Jul;128(7):1406-1407. doi: 10.1016/j.clinph.2017.02.026. Epub 2017 Mar 16. PMID: 28495216; PMCID: PMC9359665.
6. Bianchi MT, Thomas RJ, Westover MB. An open request to epidemiologists: please stop querying self-reported sleep duration. *Sleep Med*. 2017 Jul;35:92-93. doi: 10.1016/j.sleep.2017.02.001. Epub 2017 Feb 17. PMCID: PMC9371612.

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Non-Peer-reviewed Articles

1. Westover MB, O'Sullivan JA. "Towards an information theoretic framework for object recognition." IEEE 2004 International Symposium on Information Theory, Chicago, Illinois, June 27-July 2, 2004.
2. Westover MB, O'Sullivan JA. "Achievable rates for pattern recognition: Binary and Gaussian cases." IEEE 2005 International Symposium on Information Theory, Adelaide, Australia, Sept 4-9, 2005.
3. O'Sullivan JA, Singla N, and Westover MB. "Successive Refinement for Pattern Recognition," IEEE Information Theory Workshop, Punta del Este, Uruguay, Feb 2006.
4. Liddle S, Zhang R, Grover L, Khitrov M, Brown J, Cobb JP, Goldman J, Chou J, Westover MB, Yagoda D, Reisner A, "Safety Evaluation of a Medical Device Data System". International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC'12), San Diego, California, USA on August 28 - September 1, 2012.
5. Bianchi MT, Thomas RJ, Westover MB. "Response." *Sleep Med*. 2017 Aug 3. pii: S1389-9457(17)30316-7. doi: 10.1016/j.sleep.2017.07.017. [Epub ahead of print] PMID: 2884338
6. Amorim E, Zheng WL, Ghassemi MM, Aghaeiaval M, Kandhare P, Karukonda V, Lee JW, Herman ST, Sivaraju A, Gaspard N, Hofmeijer J, van Putten MJAM, Sameni R, Reyna MA, Clifford GD, Westover MB. "The International Cardiac Arrest Research (I-CARE) Consortium Electroencephalography Database." *medRxiv*. 2023 Aug 28:2023.08.28.23294672. doi: 10.1101/2023.08.28.23294672. Preprint. PMID: 37693458.
7. Sun H, Ganglberger W, Westover MB, Thomas RJ. "Sleep as a Window into Brain Health: Clinical Implications from the Sleeping Brain Wave-Structure Associations." *SLEEP*. 2025 May. (Editorial).**

8.

Book Chapters

1. Westover MB, Thomas TT, Greer DM, "Delirium" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 86-89.
2. Westover MB, Choi E, Awad KM, Greer DM, Editors. *Pocket Neurology*. Lippincott Williams & Wilkins, 2010.
3. Westover MB, Peters JM, Bromfield EB, "Seizures and Other Spells" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 57-79.
4. Westover MB, Xia, Z, Cho TA, "Meningitis, Encephalitis, & Brain Abscesses" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 136-149.
5. Thomas TT, Westover MB, "Rapidly Progressive Dementia" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 89-96.
6. Kummer TT, Westover MB, "Central Nervous System Vasculitis" in *Pocket Neurology*, ed. Westover MB, Choi E, Awad KM, and Greer DM, Lippincott Williams & Wilkins, 2010, 175-177.
7. Westover MB, Awad K, "Ischemic Stroke, Muscular Dystrophy and Friedreich's Ataxia" in *MGH Cardiology Board Review*. Springer-Verlag, London, 2013.

8. Westover MB, Edlow BL, Greer DM, “Coma After Cardiac Arrest: Management and Neurological Prognostication” in MGH Cardiology Board Review. Springer-Verlag, London, 2013.
9. Westover MB, Shafi MM, “Seizures” in Pocket ICU, Lippincott Williams & Wilkins, 2013.

VI. GRANT FUNDING

Current:

09/2021 – 05/2026	Funder: NIH-NINDS (R01 AG073410) Title: Comparative Safety of Seizure Prophylaxis within the Medicare Program Role: Co-Investigator (PI: Lidia Maria Moura)
05/2021 – 04/2026	Funder: NIH-NINDS (R01 NS120947) Title: Establishing a Brain Health Index from the Sleep Electroencephalogram Role: PI
09/2022 – 08/2027	Funder: NIH-NHLBI (R01 HL161253) Title: Data-Driven Sleep Biomarkers of Brain Health, Heart Health, and Mortality Role: PI
08/2022 – 04/2027	Funder: NIH-NIA (R01 AG073598) Title: Low Neurophysiologic Resistance to Anesthetics as a Marker of Preclinical/Prodromal Alzheimer’s Disease and Neurovascular Pathology, Delirium Risk and Inattention Role: Co-Investigator (PI: Miles Berger, via Duke subcontract)
01/2023 – 12/2027	Funder: NIH-NINDS (R01 NS126282) Title: Prospective Validation of Neurophysiologic Outcome Prediction in Acute Brain Injury Role: PI
09/2023 – 05/2028	Funder: NIH-NINDS (R01 NS130119) Title: Late-Onset Unexplained Epilepsy as a Risk Factor for Dementia Role: Co-PI (with Alice D. Lam)
12/2023 – 11/2028	Funder: NIH-NINDS (R01 NS131347) Title: Comparative Effectiveness of EEG-Guided Anti-Seizure Treatment in Acute Brain Injury Role: Co-Investigator (PI: Sahar F. Zafar)
11/2023 – 10/2028	Funder: NIH-NIA (R01 AG078261) Title: Delirium Dynamics: Understanding Causes and Effects Role: Co-Investigator (PI: Eyal Kimchi, via Northwestern subcontract)
09/2023 – 09/2025	Funder: NIH-NIA (R21 AG085495) Title: Development of Novel Polysomnography-Based Digital Biomarkers to Predict Alzheimer’s Disease and Parkinson’s Disease Role: Co-Investigator (PI: Yue Leng, via UCSF subcontract)
02/2025 – 02/2026	Funder: Takeda (PO8001791544) Title: Analysis of Risk of Narcolepsy through Clinical Notes

Role: PI

Submitted:

12/2025 – 11/2030

Funder: NIH (Pending)

Title: Leveraging Clinical Data and Automation to Predict Epilepsy After TBI

Role: Co-Investigator (PI: Jennifer Kim)

12/2025 – 11/2030

Funder: NIH (Pending)

Title: De-Biasing Effect Assessment of Seizure Effects on Neurologic Outcomes (DEBIAS)

Role: Co-Investigator (PI: Adrian Dalca)

12/2025 – 11/2030

Funder: NIH-NINDS (Pending)

Title: Automating Critical Care Electroencephalography Using AI Foundation Models

Role: PI

12/2027 – 11/2030

Funder: NIH (Pending)

Title: *Tana Mu Na?* Feasibility of Mobile EEG and Automated Intelligence to Diagnose Epilepsy in Children in Guinea-Bissau

Role: Co-Investigator (PI: Farrah Mateen)

07/2025 – 06/2030

Funder: NIH-NINDS (1UG3NS143749-01)

Title: Continuous Monitoring of Pain Using Facial Expressions in the ICU Environment (COMPANION)

Role: Co-Investigator (PI: Hao Deng)

Completed:

2012 – 2014

Funder: American Brain Foundation Clinical Research Training Fellowship

Title: Early Detection of Delayed Cerebral Ischemia After Subarachnoid Hemorrhage Using Context-Sensitive Computational EEG

Role: PI

2014 – 2019

Funder: NIH-NINDS (K23 NS090900)

Title: Quantitative Neurophysiological Monitoring and Control of Sedation-Analgesia in the ICU Environment

Role: PI

2017 – 2019

Funder: SAGE Therapeutics

Title: Automated Detection of Seizure-Spectrum Patterns and Quantitative Neurological Outcome Prediction

Role: PI

2018 – 2019

Funder: Partners Healthcare Innovation

Title: Artificial Intelligence-Based Seizure Detection and Classification

Role: PI

2018 – 2019

Funder: MGH-MIT Grand Challenge

Title: Big Data and Deep Learning for the Interictal-Ictal-Injury Continuum

Role: PI

- 2018 – 2021 Funder: Glenn Foundation for Medical Research
Title: Electroencephalogram-Based Brain Age and Its Relationship with Cognitive Function and Sleep Quality
Role: PI
- 2018 – 2023 Funder: NIH-NINDS (R01 NS102190)
Title: Investigation of Sleep in the Intensive Care Unit
Role: PI
- 2019 – 2024 Funder: NIH-NINDS (R01 NS107291)
Title: Deep Learning for the Ictal-Interictal-Injury Continuum
Role: PI
- 2018 – 2023 Funder: NIH-NINDS (R01 NS102574)
Title: Multimodal Network Connectivity Architecture (MOCA) of the Brain and Its Role in Recovery of Consciousness in Comatose Cardiac Arrest Patients
Role: Co-Investigator (PI: Ona Wu)
- 2019 – 2024 Funder: NIH-NIA (R01 AG060987)
Title: Treating Hyperexcitability in Alzheimer’s Disease with Levetiracetam to Improve Brain Function and Cognition
Role: Co-Investigator (PI: Mouhsin Shafi)
- 2019 – 2024 Funder: NIH-NIA (RF1 AG064312)
Title: Integrated Motor Activity Biomarker for Risk of Alzheimer’s Dementia
Role: Co-Investigator (PI: Kun Hu)

VII. PROFESSIONAL ACTIVITIES

Editorial Service

- 2010 Editor, Pocket Neurology, Lippincott Williams & Wilkins
2016 Editor, Pocket Neurology, 2nd Edition; Wolters Kluwer
2019-2024 Editorial Board Member, Journal of Clinical Neurophysiology

Ad Hoc Reviewer

- 2010- present Neurology
2010- present JAMA Neurology (formerly Archives of Neurology)
2006- present Philosophical Psychiatry
2010- present Journal of Clinical Epidemiology
2010- present BMJ Case Reports
2010- present IEEE Transactions on Information Theory
2010- present IEEE Transactions on Signal Processing
2010- present Journal of Medical Engineering
2010- present Journal of Clinical Neurophysiology
2010- present Journal of Neuroscience
2010- present European Journal of Neurology
2010- present Critical Care Medicine
2010- present Annals of Clinical and Translational Neurology
2010- present Journal of Neural Engineering
2010- present Journal of Cerebral Blood Flow & Metabolism
2010- present Computers in Biology and Medicine

2010- present	Proceedings of the National Academy of Sciences (PNAS)
2010- present	Brain
2010- present	Epilepsy Research
2010- present	American Journal of Respiratory and Critical Care Medicine
2010- present	SLEEP

Service as Grant Reviewer

2018-2020	Ad Hoc Member, Acute Neural Injury & Epilepsy (ANIE), NIH
2021- 2025	Permanent Member, Acute Neural Injury & Epilepsy (ANIE), NIH

University Administrative Service

2011-2017	Critical Care Informatics Committee Massachusetts General Hospital
2011-2022	Neurocritical Care Clinical Review Committee Massachusetts General Hospital
2011-2022	Co-director, MGH Neuromonitoring Committee Massachusetts General Hospital
2010-2016	Member, MGH EEG Reporting System Redesign Committee, MGH
2013-2017	MGH Infusion Pump Committee Massachusetts General Hospital
2014-2022	MGH Cooling Committee Massachusetts General Hospital
2015-2018	Member, PhD Thesis Committee (Candidate: Mohammad Ghassemi), MIT
2015-2019	Member, PhD Thesis Committee (Candidate: Jing Zhi An), MIT
2017-2019	Chair, Data Safety Monitoring Board (DSMB) for R01 (PI: Ken Solt), MGH
2018-2020	Member, DSMB for R01 (PI: Alvaro Pascual Leone), BIDMC

Service to Professional Organizations

2011-2014	TRENDs (Treatment of Recurrent Nonconvulsive SeizureS) Clinical Trial Investigators Committee, Duke University / Duke Clinical Research Institute (DCRI)
2015-2017	Member at Large Critical Care EEG Monitoring Research Consortium (CCEMRC)
2016	Co-Chair, 7th International Conference on Extreme Learning Machines (ELM2016). Singapore, December 13-15, 2016.
2017-2019	Vice-Chair, Critical Care EEG Monitoring Research Consortium (CCEMRC)
2017	Co-Chair, The 8th International Conference on Extreme Learning Machines (ELM2017). Yantai, China, October 4-7, 2017.
2019-2021	Chair, Critical Care EEG Monitoring Research Consortium (CCEMRC)
2025-2029	Chair, International League Against Epilepsy (ILAE) – Commission on Big Data and AI
2025-	Committee Member, NINDS Common Data Elements (CDE) in Epilepsy Research Oversight Committee.

Community Service

2012 Christ Lutheran Nursery School, Belmont, MA	Educational Speaker
2014 Boy Scout Troop 66, Belmont, MA. Belmont, MA	Educational Speaker
2014 Butler Elementary School, Belmont, MA,	Educational Speaker
2016 Butler Elementary School, Belmont, MA,	Educational Speaker

VIII. INVITED PRESENTATIONS (SINCE 2020)

Grand Rounds:

1. “Little; Yellow; Different: Hyperbilirubinemia in preterm infants of lactating depressed female Green Monkeys receiving CACKLE-modified dairy supplements.” Neonatology Grand Rounds, Stanford University, Stanford, CA (2022)

2. "Monitoring and Closed-Loop Control of Burst Suppression." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2013)
3. "Continuous Brain Monitoring in Critical Care." Anesthesia, Critical Care and Pain Medicine Grand Rounds, Massachusetts General Hospital, Boston, MA (2013)
4. "Why Brains Need Computers." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2016)
5. "Towards Personalized Critical Care Brain Monitoring." Neurology Grand Rounds, Stanford University, Palo Alto, CA (2017)
6. "Machine Learning in Neurology." Neurology Grand Rounds, University of Wisconsin, Madison, WI (2019)
7. "Early Ischemia Detection Using EEG in Subarachnoid Hemorrhage." Neurology Grand Rounds, University Hospitals Cleveland Medical Center, Cleveland, OH (2019)
8. "Automating Clinical Neurophysiology." Neurology Grand Rounds, Stanford University, Palo Alto, CA (2019)
9. "AI in Clinical Neurophysiology." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2020)
10. "Covid-19 Outcome Risk Prediction." Neurology Grand Rounds, Massachusetts General Hospital, Boston, MA (2020)
11. "Automated EEG Analysis in Clinical Care." Neurology Grand Rounds, University of California San Diego, San Diego, CA (2020)
12. "Denoising the Diagnosis of Epilepsy." Neurology Grand Rounds, Einstein Medical Center Philadelphia, Philadelphia, PA (2022)
13. "Quantifying the Ictal-Interictal-Injury Continuum." Neurocritical Care Grand Rounds, University of Washington, Seattle, WA (2022)
14. "Superhuman Management of Seizures in the ICU." Neurology Grand Rounds, Yale School of Medicine, New Haven, CT (2023)
15. "Optimizing Detection and Treatment of Nonconvulsive Status Epilepticus." Geschwind Pan-Harvard Grand Rounds, Harvard Medical School, Boston, MA (2023)
16. "Big Data and AI to Achieve Superhuman Detection and Treatment of Seizures in ICU Patients." Healthcare Delivery Science Grand Rounds, Beth Israel Deaconess Medical Center / Harvard Medical School, Boston, MA (2023)
17. "Toward Optimal Treatment of Harmful Brain Activity in Critical Illness." Neurology Grand Rounds, University Hospital Cleveland Medical Center, Cleveland, OH (2024)
18. "Reducing Noise in the Diagnosis of Epilepsy." Neurology Grand Rounds, University of Texas Houston, Houston, TX (2024)
19. "Reducing Noise in the Diagnosis of Epilepsy." Neurology Grand Rounds, University of Pennsylvania, Philadelphia, PA (2024)
20. "Reducing Diagnostic Noise in Neurology: Lessons from Epilepsy." Neurology Grand Rounds, Beth Israel Deaconess Medical Center / Harvard Medical School, Boston, MA (2025)
21. "Reducing Noise in Neurology: Lessons from Epilepsy." Neurology Grand Rounds, University of Michigan, Ann Arbor, MI (2025)

Local Invited Presentations:

1. "What is monkey cortex layer four for?." Neuroanatomy & Neurobiology Department, Washington University School of Medicine, St. Louis, MO (2002)
2. "EEG Patterns in the Neurological ICU." MGH Neurosciences ICU, Boston, MA (2010)
3. "Should Statins Be Avoided After Intracerebral Hemorrhage? Implications of SPARCL." MGH Stroke Research Center, Boston, MA (2010)
4. "Fundamentals of ICU EEG Monitoring." Nursing Skills Day, MGH Neurosciences ICU, Boston, MA (2012)

5. "Status Epilepticus and the Ictal–Interictal Continuum: Introduction to ICU EEG." MGH Critical Care Center, Boston, MA (2012)
6. "Neurological Prognostication in Coma After Cardiac Arrest." MGH Critical Care Center, Boston, MA (2013)
7. "Monitoring and Closed-Loop Control of Burst Suppression." Department of Brain & Cognitive Sciences, MIT, Cambridge, MA (2014)
8. "EEG-Related Skin Lesions: Statistical Process Monitoring & Risk Reduction." MGHfC Quality & Safety Operations Lead Team Meeting, Boston, MA (2014)
9. "Fundamentals of Continuous EEG Monitoring in Neurocritical Care; Quantitative Clinical EEG Fundamentals." MGH Neurosciences ICU & BWH/MGH Clinical Neurophysiology Fellows Didactic Conference, Boston, MA (2014)
10. "Automated Surveillance for Ventilator-Associated Pneumonia." MGH Infection Control Committee, Boston, MA (2015)
11. "The Role of Continuous EEG in Coma After Cardiac Arrest: Prognostic Value." MGH EEG Technologist Training Meeting, Boston, MA (2015)
12. "Quantitative Electroencephalography." Epilepsy Fellows Training Conference, MGH Neurology, Boston, MA (2017)
13. "CDAC: Bringing Big Clinical Data to Life." MGH Neurosurgery Faculty Meeting, Boston, MA (2018)
14. "Landmark Project #4: Clinical Time Series Data." Partners HealthCare EDMT Meeting, Boston, MA (2018)
15. "Validating the Brain Age Index: Modifying Brain Aging Through Exercise." McCance Center for Brain Health Faculty Meeting, Boston, MA (2019)
16. "Automatic Detection of Nonconvulsive Seizures." BIDMC Research Day, Boston, MA (2023)

National and Regional Meetings:

1. "Automatic Detection of Beta Amyloid in Digital Images Using Visual Texture." Alzheimer's Disease Research Center, Washington University, St. Louis, MO (2005)
2. "Early Detection of Delayed Cerebral Ischemia Using EEG." Greater Boston Epilepsy Society, Boston, MA (2015)
3. "Polysomnographic Signatures of Brain Age." Massachusetts Sleep Society, Westborough, MA (2017)
4. "Artificially Intelligent Brain Monitoring." World Medical Innovation Forum, Boston, MA (2019)
5. "Expert-Level Automatic Detection of Epileptiform Discharges." American Neurological Association, St. Louis, MO (2019)
6. "EEG Monitoring for Early Detection of Cerebral Ischemia." American Clinical Neurophysiology Society Annual Meeting, New Orleans, LA (2020)
7. "Automating Clinical Neurophysiology." Duke Comprehensive Epilepsy Center Quarterly Research Symposium, Durham, NC (2020)
8. "Big Data and Machine Learning to Advance Status Epilepticus Research." American Neurological Association, Virtual Meeting (2020)
9. "Automated Detection of Seizures and Ictal–Interictal Continuum Patterns Using Deep Learning." American Clinical Neurophysiology Society Annual Meeting, Virtual (2021)
10. "Sleep as a Window into Brain Health." Harvard Sleep Grand Rounds, Virtual (2021)
11. "Avoiding Pitfalls in EEG Interpretation." American Academy of Neurology Annual Meeting, Seattle, WA (2022)
12. "Artificial Intelligence in Sleep Medicine." American Neurological Association Annual Meeting, Chicago, IL (2022)
13. "Addressing Problems in Critical Care Neurophysiology Using Machine Learning." Machine Learning in Medicine Seminar Series, Carnegie Mellon University, Pittsburgh, PA (2022)
14. "Epileptiform and Non-Epileptiform Findings in ICU Electroencephalography." American Academy of Neurology Annual Meeting, Boston, MA (2023)

15. "Neurophysiological Grading of Encephalopathy." American Academy of Neurology Annual Meeting, Boston, MA (2023)
16. "Identifying and Overcoming Barriers for Clinical Use of Artificial Intelligence in Sleep Medicine." American Academy of Sleep Medicine Annual Meeting, Indianapolis, IN (2023)
17. "Optimizing Treatment of Nonconvulsive Status Epilepticus." American Epilepsy Society Annual Meeting, Orlando, FL (2023)
18. "Collaborating with Industry." American Clinical Neurophysiology Society Annual Meeting, Orlando, FL (2024)
19. "Quantitative EEG." American Clinical Neurophysiology Society Annual Meeting, Orlando, FL (2024)
20. "The EEG Map of Everything (MOE)." Critical Care EEG Monitoring Research Consortium Meeting, Orlando, FL (2024)
21. "AI in ICU EEG." American Epilepsy Society Annual Meeting, Los Angeles, CA (2024)
- 22.

International Meetings:

1. "Opportunities for Intelligent Computing in Clinical Neurophysiology." Nanyang Technical University Symposium on Novel Technologies for Clinical Practice in Neurology, Singapore (2012)
2. "Real-Time Measurement and Closed-Loop Control of Burst Suppression." IEEE Engineering in Medicine and Biology Society Conference, Osaka, Japan (2013)
3. "Optimizing the Treatment of Refractory Status Epilepticus." Networks in Epilepsy Conference, Beijing, China (2015)
4. "Engineering in Neurocritical Care: Modeling & Control of Refractory Status Epilepticus." International Symposium on EEG Signal Analysis, Xi'an, China (2015)
5. "Quantitative Electroencephalography for Detection of Seizures and Ischemia." Beijing ICU Neuroelectrophysiological Monitoring Forum, Beijing, China (2015)
6. "Why Brains Need Computers: How Machine Learning and Big Data Can Help Neurology." Extreme Learning Machines Annual Conference, Singapore (2016)
7. "Machine Learning in Neurocritical Care." Northwestern University International Forum on Big Data in Healthcare, Xi'an, China (2017)
8. "Quantitative EEG: Detecting Ischemia and Seizures." International Congress of Clinical Neurophysiology, Washington, DC (2018)
9. "Removing the Confusion from Toxic-Metabolic Encephalopathy." University of Twente Mini-Symposium, Enschede, Netherlands (2018)
10. "Mapping the Ictal–Interictal–Injury Continuum." Neurocritical Care Society Annual Meeting, Vancouver, Canada (2019)
11. "Superhuman Detection of Seizures, Rhythmic and Periodic Patterns in ICU EEG." International Congress on Clinical Neurophysiology (ICCN), Geneva, Switzerland (2022)
12. "Quantifying the Ictal–Interictal Continuum." International Conference on Technology and Analysis for Seizures (ICTALS), Bern, Switzerland (2022)
13. "Denoising the Diagnosis of Epilepsy." AI in Epilepsy Conference, Breckenridge, CO (2025)
14. "MORGOTH: Automating the Interpretation of Clinical Neurophysiology Tests." International Conference on Technology and Analysis for Seizures (ICTALS), Montreal, Canada (2025)
15. "Data-Driven Biomarkers for Brain Health from Sleep." SLEEP 2025 Annual Meeting, Seattle, WA (2025)
16. "Artificial Intelligence in Neurology: Promise and Peril." Donald Baxter Lecture, Montreal Neurological Institute, Montreal, Canada (2025)

IX. TEACHING

Teaching during Stanford appointment:

None.

Teaching prior to Stanford:

Courses

2001–2002	Instructor for Computational Neuroscience, Washington University School of Medicine, Department of Anatomy and Neurobiology.
2003	Instructor for Detection and Estimation Theory, Washington University, Department of Electrical and Systems Engineering.
2004	Instructor for Information Theory, Washington University, Department of Electrical and Systems Engineering.
2008–2009	Instructor for Neuroanatomy Laboratory, Harvard Medical School.
2012–2025	Lecturer for Medical Decision Analysis and Diagnostic Test Interpretation (HST-192.0), Harvard–MIT Program in Health Sciences and Technology.
2013–2017	Course Director and Lecturer for Medical Decision Analysis and Diagnostic Test Interpretation (HST-192.0), Harvard–MIT Program in Health Sciences and Technology.

Tutorials and Lectures

10/2009	Granger Causality Analysis Techniques for EEG Data. Clinical Neurophysiology Research Group, MGH, Boston, MA.
5/2010	Synchronization Likelihood for Functional Connectivity Analysis of EEG Data. Clinical Neurophysiology Research Group, MGH, Boston, MA.
7/2008	Probability in Medical Reasoning. Neurology Residents, Brigham and Women's Hospital, Boston, MA.
9/2009	Invited lecturer, Bayes' Rule for Physicians. Partners Neurology, MGH/BWH, Boston, MA.
6/2010	EEG Alphabet Soup: GCSE, NCSE, PLEDs, BIPLEDs, GPEDs, & DEADs. Partners Neurology, MGH/BWH, Boston, MA.
8/2011	Status Epilepticus: Diagnosis and Management. Partners Neurology, MGH/BWH, Boston, MA.
9/2012	Introduction to ICU EEG: Seizures, Spikes, PEDs, Ischemia, and Coma. Partners Neurology, MGH/BWH, Boston, MA.
7/2018	Quantitative Electroencephalography. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.
7/2019	Recognizing Seizures and Ischemia in Critical Care EEG Monitoring. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.
6/2020	Emergency Electroencephalography. Emergency Medicine Residents, Massachusetts General Hospital, Boston, MA.
7/2020	Introduction to Spectrograms in ICU EEG. Neurophysiology and Epilepsy Fellows, MGH Epilepsy Division, Boston, MA.

Clinical Supervisory and Training

2012–	Supervisor, MGH Adult Epilepsy Clinic (½-day session per week).
2012–	Attending Physician, MGH Inpatient Neurology Service (2-week block per year), supervising neurology residents and Harvard medical students.
2012–	Attending Physician, MGH General Neurology Consult Service (2-week block per year), supervising neurology residents.
2012–	Director, MGH Critical Care EEG Monitoring Service (≈10 weeks per year), supervising Epilepsy and Clinical Neurophysiology Fellows.
2012–	Attending, MGH Epilepsy Monitoring Unit (≈6 weeks per year), supervising Epilepsy and Clinical Neurophysiology Fellows.
2012–	Faculty, Outpatient EEG Reading Service (≈2 days per month), supervising Epilepsy and Clinical Neurophysiology Fellows.

Research Supervisory and Training Responsibilities

2012– Supervisor of postdoctoral research fellows (average 5–10 fellows per year), Massachusetts General Hospital.

2012– Supervisor of graduate students (average 2–4 per year), Massachusetts General Hospital and MIT.

X. TRAINEES

Postdoctoral Fellows

2013-2014 Jing Jin, PhD from NTU in Singapore. postdoctoral fellow at MGH

2013-2014 Zhang Rui, PhD, visiting scholar from NTU in Singapore, mentoring for 2 months through present. Professor of Mathematics at Northwestern University, Xi'An, China

2014-2018 Sunil Belur Nagaraj, PhD, electrical engineering postdoctoral fellow. an Assistant Professor at University of Groningen, Department of Clinical Pharmacology, Groningen, Netherlands

2014- Lauren McClain, PhD. Research Fellow. a PhD student at Fordham University, New York

2019- Eline Oppersma, PhD. Visiting scholar. Assistant Professor of Technical Medicine at University of Twente, Netherlands

2019- WenDong Ge, PhD. Postdoctoral fellow.

2015–present Sahar Zafar, MD (K23).

2014–2020 Jennifer Kim, MD, PhD (K23).

2014–2019 Edilberto Amorim, MD (K23).

2019–present Marta Priscila Bento Fernandes, PhD.

2020–2022 Rajesh Amerineni, PhD.

2019–2022 Wendong Ge, PhD.

2019–2022 Weilong Zheng, PhD.

2022–2024 Yalda Amidi, PhD.

2021–2023 Samaneh Nasiri, PhD.

2019–present Wolfgang Ganglberger, PhD.

2024–present Shadi Sartipi, PhD.

2024–present Alexandra Tautan, PhD.

2024–present Ayush Tripathi, PhD.

2024–present Samuel Waters, PhD.

2024–present ChenXi Sun, PhD.

2024–present Charlotte Maschke, PhD.

2024–present Bruce Nearing, PhD.

2024–present Ayan Mitra, PhD.

2016–2018 Sungtae An, MS / PhD.

2016–present Haoqi Sun, MD.

2013–present Jing Jin, PhD.

Graduate Students

2014- Mohammed Ghassemi, MIT PhD candidate (now graduated), mentorship resulted in Advanced Multimodal Neuroimaging Training Program (AMNTP) fellowship award, 3 published manuscripts. CEO of a startup company, Ghamut

2014- Jingzhi An, PhD candidate at Harvard/MIT HST program.

2014-2019 Moura, Junior Valdery, MS (computer science), programmer analyst / research assistant at MGH. Project: establishing a database for the Clinical Data Animation Center (CDAC). 3 peer-reviewed publications to d. Project Manager for Landmark Project #4 at Partners Healthcare

2014-2017	Apeksha Shenoy, MS (electrical engineering), programmer analyst / research assistant. a Senior Consultant at Boston Strategic Partners
2015-	Elham Bagheri, MS (electrical and systems engineering). Nanyang Technical University (NTU), Singapore, PhD candidate, co-supervised with Prof.
2015-	John Thomas, MS (electrical and systems engineering). Nanyang Technical University (NTU), Singapore, PhD candidate, co-supervised with Prof.
2015-2016	Siddharth Biswal, MS, computer programmer / research assistant. a PhD candidate at Georgia Institute of Technology
2015-	Gladia Hotan, PhD candidate at MIT. Mentorship resulted in 2 publications to date.
2015-2016	Ellis Wickering, MS, University of Twente, Netherlands. working as an computer programmer at Prolira
2016-	Jefferson T, Oliva, PhD candidate in University of Sao Paulo, Brazil, Institute of Mathematics and Computer Sciences.
2016-	Jord Vink, MS, University of Twente, Netherlands, 3 month research internship with me at MGH, co-mentored with Dr. working at Brain Science Tools, Netherlands
2016-	Wei-Hung Sun, MD, masters student in the Harvard Biomedical Informatics program. a PhD candidate at MIT in computer science
2016-	Manohar Ghanta, MS (computer science). Computer engineer / research assistant.
2016-	Yu-Ping Shao, MS (computer science). Computer engineer / data scientist / research assistant.
2016-	Sungtae An, MS (computer science). Georgia Tech PhD candidate, co-supervised with Dr.
2016-	Kunal Malhotra, MS (computer science). Georgia Tech (GAT) PhD candidate, co-supervised with Dr.
2016-	Haoqi Sun, MD (computer science). Nanyang Technical University (NTU), Singapore, PhD candidate, co-supervised with Prof.
2016-	Senan Ebrahim, MD, PhD student in the Harvard Medical School HST program; co-supervised with Dr.
2016-2017	Sophia Bechek, research assistant. a PhD candidate at Stanford
2016-2018	Balaji Goparaju, MSc. a Data Scientist at Novartis Institutes for BioMedical Research (NIBR)
2016-	Jiang Ling Song, MS (mathematics), Northwestern University, Xian, China.
2017	Dennis Rebergen, Research internship.
2017-	Emile d'Angremont, Research internship.
2017-2018	Peiyun Ni, MD/PhD candidate at Harvard Medical School / MIT. Project: Antiepileptic drug treatment after an unprovoked first seizure: A decision analysis.
2016-2017	Nishant Sinha, PhD candidate (now graduated) at NTU, Singapore, department of computer science.
2018-	Qiang Li, PhD candidate in Mathematics, Northwestern University, Xi'An, China.
2018-2019	Kicky van Leeuwen, Masters student in Technical Medicine from U. Twente, Netherlands. a Medical Machine Learning Scientist at Medix
2018-	Michael Leone, MSc.
2014-2020	Jingzhi An, PhD.
2014-2019	Mohammad Ghassemi, PhD.
2016-2017	Jefferson T. Oliva, PhD.
2018-2020	Qiang Li, PhD.
2024-present	Arnav Gupta.
2021-2025	Thijs Nassi, MSc.

Medical Students / Residents / Fellows

	Mentorship for 4 months (2 summers), resulted in 4 scientific publications.
2011-2012	Sarah Wahlster, MD, Neurology resident. Assistant Professor of Neurology at University of Washington
2011-2012	Craig Williamson, MD, Neurology resident, Mentorship for 1 year, resulted in 1 published scientific manuscript. Assistant Professor of Neurology at University of Michigan
2013-2014	Lidia Moura, MD, Epilepsy Fellow at MGH. Two abstracts, 1 published scientific manuscript. Assistant Professor at Harvard Medical School
2013-2014	Marcus Ng, MD, Epilepsy Fellow at MGH. One published manuscript. Assistant Professor at University of Manitoba, Winnipeg, Canada
2013-2014	Deirdre O'Rourke, MD, Epilepsy Fellow at MGH, mentorship for 1 year, resulted in 1 abstract and 1 published manuscript. Neurologist at Connolly Hospital Blanchardstown/ Beaumont Hospital Dublin
2013-2014	Sandipan Pati, MD, Epilepsy Fellow at MGH. 1 abstract, 2 published scientific manuscripts. Assistant Professor of Neurology at UAB Birmingham
2013-2014	Aaron Berkowitz, MD, PhD, neurology resident at BWH/MGH/Partners. Assistant Professor of Neurology at Harvard Medical School / Brigham and Women's hospital
2014	Vincent Alvarez, MD, visiting Research Fellow from Universite de Lausanne (Switzerland). 5 published manuscripts. an Attending Neurologist at Hopital du Valais, Sion, Switzerland
2014	Sanjay Menon, MD, Neurosciences ICU Fellow at MGH, mentorship for 6 months.
2014	Ammar Kheder, MD, Epilepsy Fellow at MGH, mentorship for 1 year. an Assistant Professor Of Clinical Neurology at U
2014-	Aaron Struck, MD, Epilepsy Fellow at MGH. 2 abstracts, 8 published manuscripts. Assistant Professor of Neurology at University of Wisconsin
2014-2016	Gina Deck, MD, Epilepsy Fellow at MGH, mentorship for 1 year. an Assistant Professor of neurology at The Warren Alpert Medical School of Brown University
2014-	Edilberto Amorim, MD, neurointensive care fellow at MGH. Research on the prognostic value quantification of EEG reactivity in coma in patients with anoxic brain injury.
2014-	Jennifer Kim, MD, PhD, neurology resident at MGH/BWH. Project: "Identifying EEG Hallmarks for Increased Seizure Risk in SAH: A Quantitative EEG Analysis." Starting in July 2019, will be Assistant Professor of Neuro.
2014-2017	Katherine O'Connor, Project: Predicting delayed cerebral ischemia in patients with aneurysmal subarachnoid hemorrhage.
2015-	Sahar Zafar, MD, neurointensive care fellow at MGH. Research on the impact of pathological brain activity and its treatments on neurological outcome in ICU patients.
2015-2017	Carlos Muniz, MD. Epilepsy / Neurophysiology Fellow. a practicing neurologist in Towson, Maryland
2016-	Ilya Toloch, MD, PhD, neurology resident at BWH/MGH/Partners.
2016-2016	Eva Postma, MD, University of Twente, Netherlands, visiting scholar for 6 months in my laboratory, co-mentored by Dr.
2016-2017	Jia Jian, PhD (mathematics), Northwestern University (NWU), Xian, China. an Associate Professor of Mathematics at NWU
2016	Honghui Zhang, PhD (mathematics), Northwestern University, Xian, China.

2014-2018	Durga Jonnalagadda, MD, Projects: Diagnosing severity of delirium in hospitalized patients from EEG recordings.
2017-	Aline Herlopian, MD. Fellow in Clinical Neurophysiology and Epilepsy at Harvard Medical School, MGH. Projects: Characterizing EEG patterns in patients with CAR-T cell encephalopathy, and Measuring expert inter-rater a. Assistant Professor of Neurology at Yale
2017-	Hassan Aboul Nour, MD. Project: Estimating the effect of exposure to inadvertent pharmacological burst suppression during critical illness on neurologic outcomes.
2017-2018	Elahe Bordbar, MD. Research Fellow. a Resident in Neurology at Temple University
2017-	Muhammad Edhi, MD. Research Fellow.
2017-	Farrukh Javed, MD. Research Fellow. a Neurology Resident at University of Virginia
2017-	Solomon Kassa, MD. Research Fellow.
2017-2019	Maryum Shoukat, MD. Research Fellow.
2017-	Mohammad Tabaeizadeh, MD. Research Fellow.
2017-2018	Erik Bao, MD candidate in the Harvard/MIT Health Sciences & Technology.
2017-2018	Monica (Ling-Ya) Chao, MD candidate in Harvard HST program.
2018-	Kendrick Shaw, MD, PhD. Anesthesia Resident and incoming faculty (Instructor) at Harvard Medical School / MGH. Project: Predicting delirium using electronic health record data.
2018-	Chris McGray, MD, PhD. Fellow in Epilepsy and Clinical Neurophysiology.
2018-	Sophia Ryan, MD. Harvard Medical School/CRICO Patient Safety and Quality Fellow at Harvard Medical School.
2018-	Benjamin Grannan, MD. Resident in Neurosurgery at Harvard Medical School / MGH. Project: Deep learning analysis of CT images to detect cases in need of surgical intervention.
2019-	Eliezer Sternberg, MD. Fellow in Epilepsy and Clinical Neurophysiology.
2014–2015	Elliott Akama-Garren.
2014–2020	Jennifer Kim, MD, PhD.
2018–present	Eyal Kimchi, MD, PhD (K23).
2022–present	Daniel Goldenholz, MD, PhD (K23).
2022–present	Peter Hadar, MD, PhD.
2022–2025	Wan Yee Kong, MD, MSc.
2022–present	James Lucarelli, MD, PhD.
2023–present	Callison Alcott, MD, PhD.
2011–2012	Sarah Wahlster, MD.
2011–2012	Craig Williamson, MD.
2013–2014	Lidia Moura, MD.
2013–2014	Marcus Ng, MD.
2013–2014	Patrick Chen, MD.
2013–2014	Deirdre O’Rourke, MD.
2013–2014	Sandipan Pati, MD.
2013–2014	Ammar Kheder, MD.
2013–2014	Aaron Berkowitz, MD, PhD.
2017–2019	Luis Paixao, MD.
2017–present	Aline Herlopian, MD.
2020–2021	Nitish Harid, MD.

Undergraduate Students

2010; 2011 Nathaniel Eiseman, Senior at Tulane.

2012	Matthew Ferguson, sophomore at Washington University in Saint Louis.
2012	Justine Cormier, BS, medical student at Boston University School of Medicine. Neurology Resident at Yale
2014	Elliot Akama-Garren, MIT undergraduate, mentorship for 6 months; resulted in 1 publication. an MD-PhD student at Harvard Medical School
2016-	Greta Gadbois, BS (mathematics), research assistant.
2017	Wade Whitt, undergraduate student at BYU. Projects: Correlation of EEG findings in delirious patients with neurologic outcomes.
2018-	Jacob Hogan, undergraduate student at BYU. Project: Estimating the causal effect of burst suppression on neurologic outcomes.
2018–2023	Alek Westover.
2021–2022	Danny Jones.
2021–2022	Jaden Barfuss.
2021–2023	Jayme Banks.
2023–2024	Leia Jekel.
2023–2024	Rebecca Milde.
2023–2024	Moritz Alkofer.
2023–2024	Elijah Davis.
2023–present	Dan Sumsion.
2023–present	Lydia Petersen.
2023–present	Kiwi (Yiwen) Xiong.
2024–present	Luke Jackson.
2024–present	Carlos Nuñez-Sosa.
2024–present	Amber Peng.
2024–present	Mitchell McCauley.
2024–present	Patrick Wynn.
2024–present	Elijah Keldsen.

High School Students

2018	Alek Westover, High school student.
2016–2017	Sophia Bechek.
2023–2024	Samvrit Rao.